

Candidate Seat No: \_\_\_\_\_  
**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**Master of Science (MSC-IT)**  
**Semester-I University Examination MARCH-2018**  
**MS110 Database Technologies**

**Date: 26.03.18, Monday**  
**Time: 10:00 Am to 1:00 Pm**

**Maximum Marks: 70**

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**SECTION-1**

**Q.1 Do as directed. [11]**

1. \_\_\_\_\_ is not a state of transaction.

A. Active              B. Wait              C. Aborted              D. Committed

2. The SQL keyword \_\_\_\_\_ is used with wildcards.

A. LIKE              B. IN              C. NOT IN              D. IN AND NOT IN

3. One database contains one Physical level. **State True\False.**

4. INSERT, UPDATE AND DELETE are \_\_\_\_\_ commands.

A. DDL              B. DCL              C. DML              D. TCL

5. Atomic property of transaction means "All or None". **State True\False**

6. Deadlock is 'system type of error'. **State True\False.**

7. \_\_\_\_\_ type of lock can be acquired by transaction.

A. Shared              B. Exclusive              C. Both (A) and (B)              D. None

8. Managing lock on data items is responsibility of \_\_\_\_\_.

A. User              B. Recovery manager              C. Lock Manager              D. Query manager

9. Forward recovery is also called Roll forward. **State True\False.**

10. What is Relation?

11. Define Database Management System.

**Q.2 [A] Explain SELECT command with example. [06]**

**[B] Write short note on DBA. [06]**

**OR**

**Q.2 [A] Explain ER Model with example [06]**

**[B] Explain Three Level Architecture Objectives of database. [06]**

**Q.3 [A] Explain ACID property of transaction with example. [06]**

**[B] Explain Equivalent schedule and Serial schedule with example [06]**

**OR**

**Q.3 [A] What is deadlock? Explain with example [06]**

**[B] Write short note on Database Recovery. [06]**

## SECTION-2

**Q.4 Do as directed**

**[11]**

1. Write Full form of ORDBMS \_\_\_\_\_.
2. Select query in OQL return objects or collection of objects .**State True\False**
3. Limitation of \_\_\_\_\_ is Low performance in web application.  
A. DBMS                      B. RDBMS                      C. ODBMS                      D. ORDBMS
4. in DDBMS D mean \_\_\_\_\_.  
A. Develope                      B. Describe                      C. Distributed                      D. Double
5. Encapsulation is the ability of an object to take on many forms. **State True\False**
6. Create table Emp\_Dept as Select \* from emp where Dname='Production' ;  
What type of fragmentation is in above query?  
A. Vertical                      B. Horizontal                      C. Both (A) and (B)                      D. None
7. Oracle's catalog is called \_\_\_\_\_.  
A. Database                      B. Data Dictionary                      C. Metadata                      D. None
8. \_\_\_\_\_ is a technique to finding the cheapest evaluation plan for a query.  
A. Optimization                      B. Parsing                      C. Evaluation                      D. None
9. \_\_\_\_\_ is used to represent a relational algebra.  
A. Query Graph                      B. Query Tree                      C. Query plan                      D. Query Manager
10. XSLT is a language designed for translation from XML to HTML **True/False.**
11. Shared-Memory system share single Main memory. **State True\False**

**Q.5 [A]** Explain Shared Disk and Shared Nothing architecture of parallel database. **[06]**

**[B]** Write detail note on Applications of Data mining. **[06]**

**OR**

**Q.5 [A]** Explain XML with advantages and example. **[06]**

**[B]** Write short note on STAR Schema of Data warehouse. **[06]**

**Q.6 [A]** Write detail note on Active database. **[06]**

**[B]** Write detail note on Multimedia database. **[06]**

**OR**

**Q.6 [A]** Write detail note on GIS **[06]**

**[B]** What is Mobile database? Explain Peer to Peer mobile database in detail **[06]**